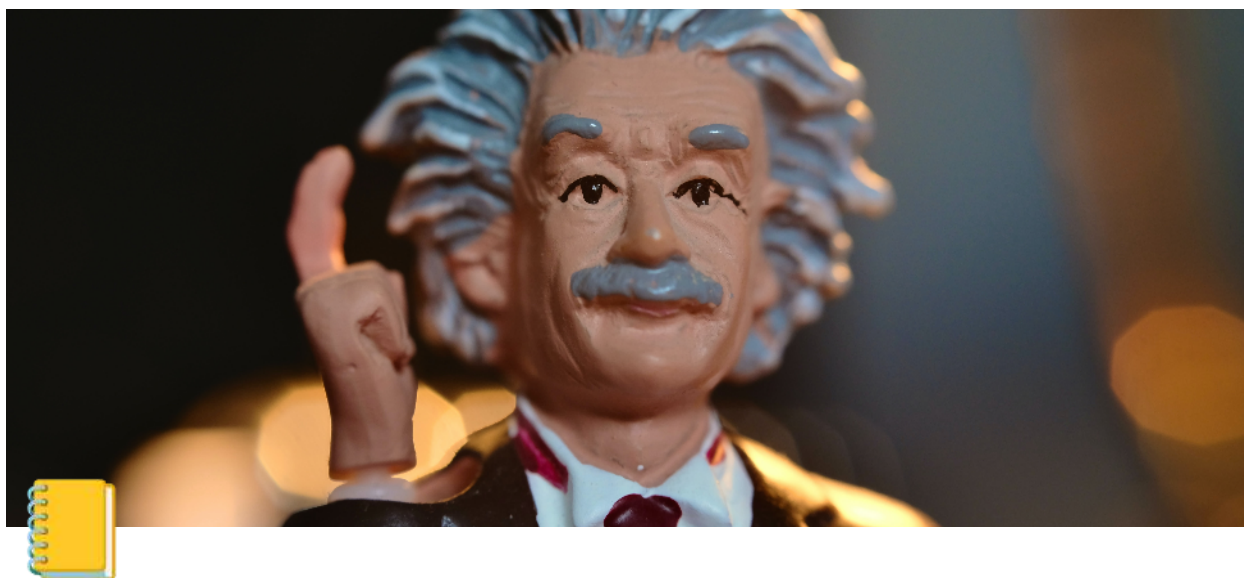


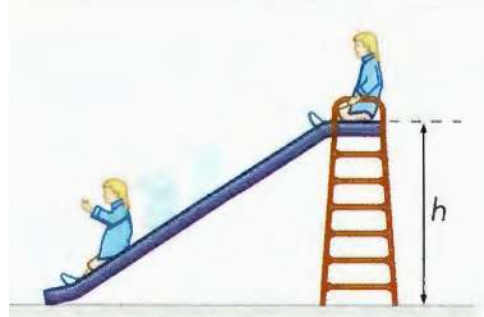


Physics Notes By Vasumitra Gajbhiye

8. Work and Power

- Work done = Energy transfer
- Mechanical or electrical work is equal to energy transferred.
- Work done = force x distance **moved by the force in the direction of the force.**
- $W = Fd = \Delta E$
- $1J = 1Nm$
- In the example on the right, To calculate the work done by gravity, we need to know the vertical distance, h , because this is the distance moved in the direction of the force. If we calculated the work done

as weight x distance moved down the ramp, we would get an answer that was too large.



- To increase power :
 - Increase the mass of the object, time & distance = constant
 - Increase the distance of motion, time & mass = constant
 - Decrease the time for work done (do the work faster), force & distance = constant.
- Power = work done / time take = energy transferred / time taken
- $p = W/t = \Delta E/t$
- $1J/s = 1W$
- $1000W = 1kW$ and $1000kW = 1MW$
- % efficiency = (useful power output / power input) x 100

9. The Kinetic Particle Model Of Matter

- The distinguishing properties of the three states of matter

State	solid	liquid	gas
Size	rigid, fixed shape, fixed volume; cannot be squashed	not rigid, no fixed shape, fixed volume; cannot be squashed	not rigid, no fixed shape, no fixed volume; can be squashed
Shape	takes the shape of its container	takes the shape of its container	expands to fill its container

- Structure of solids, liquids and gases in terms of arrangement, separation and motion of the particles.

State	Arrangement and separation of particles	Motion of particles
Solid	The particles are packed closely together, in a regular pattern. Notice that each particle is in close contact with all its neighbours.	Because the particles are so tightly packed, they cannot move around. However, they do move a bit. They can vibrate about a fixed position. The hotter the solid, the more they vibrate.
Liquid	The particles are packed slightly less closely together than in a solid. The particles are arranged randomly rather than in a fixed pattern.	Because the particles are slightly less tightly packed than in a solid, they can move around. So the particles are both vibrating and moving from place to place. The hotter the liquid is, the faster its molecules move.
Gas	The particles are widely separated from one another. They are no longer in contact, unless they collide with each other. In air, the average separation between the particles is about ten times their diameter.	The particles move freely about, bouncing off one another and off the walls of the container. In air at room temperature, the average speed of the particles is about 500 m/s and this increases with temperature.

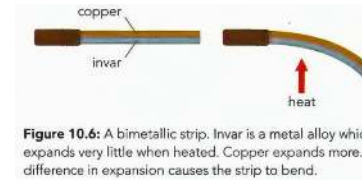
- When smoke particles are observed through a microscope:
 - **Description of motion**
 - Smoke particles show random/ haphazard/ unpredictable movement
 - Smoke particles show sudden changes of directions/ zig-zag motion
 - Smoke particles appear/ disappear from view OR go out of/ come into focus.
 - **Explanation of motion**
 - Air molecules collide with smoke particles OR smoke particles collide with / moved by air molecules.
 - Air molecules faster
 - Air molecules move randomly
 - Air molecules smaller / lighter
- Each particle of a solid is strongly bonded to its neighbours. This is because the forces between particles are strongest when the particles are close together.
- In a liquid, the particles are slightly further apart and so the forces between them are slightly weaker

- In a gas, the particles are far apart, so that the particles do not attract each other and can move freely about.
 - Gas molecules collide with the walls of the container and exert a force on the wall of the container.
 - Higher the temperature of a gas, the faster its particles are moving. The particles will hit the walls more often and with more force. This increases the pressure.
-
- Internal energy is the total energy of all of the particles.
 - Temperature is a measure of the average kinetic energy of the individual particles.
 - A bath of water at 50°C can have the same temperature as a cup of tea, but it has more internal energy than the cup of tea because it has far more molecules.
 - Celsius scale have two fixed points:
 - 0°C: the melting point of pure ice at atmospheric pressure
 - 100°C: the boiling point of pure water at atmospheric pressure.
 - $T(K) = \Theta(^{\circ}C) + 273$
 - The pressure of a gas is caused by molecules of gas hitting the walls, changing momentum and so exerting a force. The pressure is the force per unit area on the walls of the container.
 - $pV = \text{constant}$ **OR** initial pressure x initial volume = final pressure x final volume.
 - $1Pa = N/m^2$
 - $1atm = 100kPa$
-

10. Thermal Properties of Matter

- The increase in volume of a material when its temperature rises.
- Red-hot rivet is passed through holes in two metal plates and then hammered until the end rounded. As the rivet cools, it contracts pulls the two plates together tightly.

- A steel 'tyre' can be fitted on to the wheel of a train when the tyre is very hot. It then cools and contracts, so that it fits tightly on to the wheel.
- Bimetallic strip is made of two metals joined firmly together. When heated one metal expands much more than the other. The metal that expands more is on the outside of the curve, because the outer curve is longer than the inner one. Used in fire alarms and thermostats (used to control temperature of devices like ovens and irons.)



- Glass containers may crack when hot liquid is placed in them. This is because the inner surface of the glass expands rapidly, before the thermal energy has passed through to the outer surface. The force of expansion cracks the glass.
 - When gas is heated, particles move faster, push with greater force, move further apart. The gas has expanded. When gas is hot, it expands, its density decreases. This is why hot air rises.
 - Solid expands least, liquids expand more than solid, gases expand most.
 - When a material expands, particles in it have more energy, so they can move around more and take up more space.
 - It is difficult for solids to push their neighbouring atoms because of strong intermolecular forces, so solids don't expand much.
 - When a gas is heated, its particles move about more rapidly, and it is easy for them to push the walls of their container further apart, so that the gas takes up more space.
-
- Internal energy is a measure of the total energy of all the particles in the object. This includes both the kinetic energy of the particles and chemical potential energy of the bonds between them.
 - A rise in the temperature of an object increases its internal energy.
 - An increase in temperature of an object is an increase in the average kinetic energies of all of the particles in the object

- **Specific heat capacity is the energy required per unit mass per unit temperature increase**
 - Unit of specific heat capacity is $J/(kg^{\circ}C)$ and its symbol is “c”.
 - $c = \Delta E / (m\Delta\theta)$
-
- During melting and boiling energy must be provided to break bonds. Energy is also needed to overcome the attraction between the particles.
 - Melting point of pure water is $0^{\circ}C$. Boiling point of pure water is $100^{\circ}C$. At higher altitude, water boils at a lower temperature and melts at higher temperature.
 - In condensation, particles slow down and the particles move close together.
 - In solidification, or freezing, as the liquid loses energy, its particles slow down and the bonds holding the particles together re-form.
 - Not all substances melt or boil when they are heated. some burn, and others decompose into simpler substances before they have a chance to change state.
 - Evaporation is the escape of more energetic particles from the surface of a liquid.
 - During evaporation, most energetic molecules escape from the surface of the liquid, leaving the less energetic molecules in the liquid, the average energy of the remaining particles is less, so liquid cools down.
 - When a liquid evaporates in contact with a surface (e.g. skin), molecules gain kinetic energy, most energetic molecules evaporate from the surface, taking the energy from evaporation from the surface. This cools down the surface.
 - Boiling happens at a precise temperature. Evaporation occurs at all temperature
 - Boiling happens throughout the liquid. Evaporation only happens at the surface
 - A boiling liquid bubbles. a liquid can evaporate without bubbles.

To increase the rate of evaporation:

1. Increase the temperature → as the temperature increase molecules gain kinetic energy. Greater average kinetic energy. More of the particles will have enough energy to escape. So rate of evaporation increase.
2. Increase the surface area → more of the particles are close to the surface, and so they can escape more easily. So rate of evaporation increase.

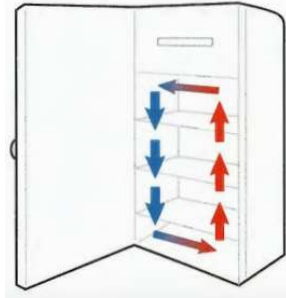
3. Blowing air across the surface → Draught is moving air. When particles escape from the water, they are blown away so that they cannot fall back in to the water. So rate of evaporation increase.
-

11. Thermal Energy Transfers

- When a rod is heated, the atoms are vibrating much more at the hot end than they are at the cold end. As the atoms vibrate, they collide with their neighbours. This process results in each atom sharing its energy with its neighbouring atoms. The collisions gradually transfer energy from the atoms at the hot end to those at the cold end.
 - In metals, electrons are free to move (they are delocalised). At the hotter end, electrons gain energy, move through the lattice, collide with distant atoms and transfer energy.
 - In liquids particles are in close contact with one another. However, as the particles are free to move, vibrations are not passed on as easily as in a solid.
 - The particles in gases are very spread out, making gases very poor conductors of thermal energy.
 - There are many solids that conduct thermal energy better than thermal insulators but do so less well than good thermal conductors.
-
- Convection is an important method of thermal energy transfer in liquids and gases.
 - Convection doesn't happen in solids as the particles are in fixed position so they cannot flow.
 - When water is heated using a Bunsen burner, water above the flame expands. Expansion means an increase in volume while the mass stays constant. this means that density decrease. So now its density is less than the that of the surrounding water, and it floats upwards. Colder water at the top, which is more dense, sinks and replaces it.
 - In convection, energy is transferred through a material from a warmer place to a cooler place

by the movement of the material itself.

- In a refrigerator, cold air sinks from the freezing compartment. If the freezer was at the bottom, cold air would remain there, and the food at the top would not be cooled.

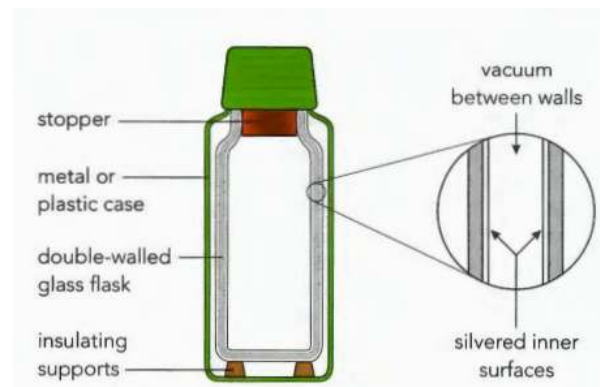


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- Thermal radiation is infrared radiation and all objects emit this radiation.
 - Thermal energy transfer by thermal radiation does not require a medium.
 - For an object to be at a constant temperature it needs to transfer energy away from the object at the same rate that it receives energy.
 - infrared radiation:
 - is produced by warm or hot objects
 - is a form of electromagnetic radiation
 - travels through empty space (and through air) in the form of waves
 - travels in straight lines
 - warms the object that absorbs it
 - is invisible to the naked eye
 - can be detected by nerve cells in the skin
 - If the rate at which an object receives energy is less than the rate at which it transfers energy away its temperature decreases.
 - If the rate at which an object receives energy is more than the rate at which it transfers energy away its temperature increase.
 - A surface that is a good reflector is a poor absorber and poor emitter.
 - A surface that is a poor reflector is a good absorber and good emitter.
 - Shiny or white surfaces are the best reflectors, so worst absorber and worst emitter.
 - Matte black surfaces are the worst reflectors, so best absorber and best emitter.

- Any object which is hotter than its surrounding (higher surface temperature) radiates more energy per second than it absorbs and so will cool down.
- Any object which is cooler than its surrounding (lower surface temperature) absorbs more energy per second than it radiates and so will heat up.
- An object with a large surface area emits thermal energy at faster rate.

- Methods of retaining energy in a house in cold climate:
 - Thick curtains, draught excluders → stops convection currents, and so prevents thermal energy transfer.
 - Loft and underfloor insulating materials → prevents conduction of thermal energy through floors and ceilings.
 - Double and triple glazing of windows → vacuum between glass panes cuts out losses or gain by conduction and convection.
 - Cavity walls → reduces thermal energy loss or gain by conduction.
 - Foam or rockwool in wall cavity → further reduce thermal energy transfer by convection.

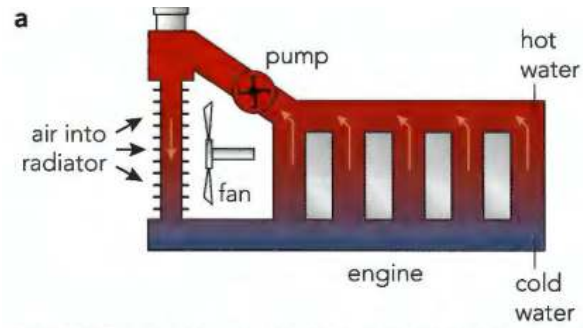
- Working of vacuum flasks:
 - Glass is generally used, because glass is a good insulator.
 - Air is removed from the gap between the double walls, creating a vacuum. This reduces losses by conduction and convection because both of them need a material to travel through.
 - The silver coating on the glass induces losses by radiation by reflecting any infrared inflation.



- The stopper is made of plastic and it prevents losses by convection and evaporation.

- Radiator of car uses the following features:

- Specific heat capacity: water flows around the block to absorb thermal energy. Water is a good choice as it has a very high specific heat capacity.
- Convection: as the water is heated, a convection current flows in the direction shown by the arrows. The pump is used to speed up this flow.
- Radiation: the fins have a large surface area and are black to increase the rate of thermal energy radiation.



- Conduction: the radiator has metal fins so the thermal energy is conducted to all parts of the radiator.

- The Sun's radiation warms the Earth. The warm Earth emits some infrared radiation. Gases in the Earth's atmosphere, such as carbon dioxide, absorb some of this thermal energy and this warms our atmosphere. This is the greenhouse effect.
- Warm air rises above the Equator, and colder air sinks in subtropical areas. This creates the pattern of trade winds that are experienced in the tropics.
- Ocean currents help to spread thermal energy from equatorial regions to cooler parts of the Earth's surface. Warm water at the surface of the sea flows towards the poles. In polar regions, colder water sinks and flows back towards the Equator. Provided this pattern remains constant, this helps to make temperate regions of the world more habitable.

12. Sound

- Sound waves are longitudinal.
- The approximate range of frequencies audible to humans as 20 Hz to 20 000 Hz.
- A medium is needed to transmit sound waves.
- The speed of sound in air is approximately 330–350 m/s.

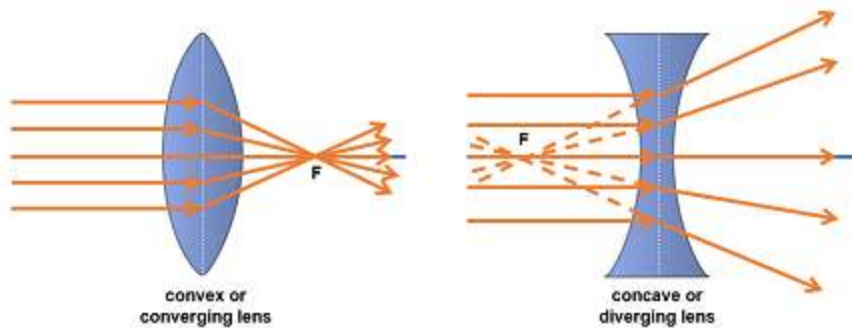
- If amplitude increase loudness increase. If amplitude decrease then loudness decrease.
 - If frequency increase pitch increase. If frequency decrease pitch decrease.
 - Echo is a reflection of sound.
 - Ultrasound is sound wave with frequency greater than 20kHz.
 - Compressions are regions of high pressure where particles are close together.
 - Rarefaction are region of low pressure where particles are far apart.
 - In general, sound travels faster in solids than in liquids and faster in liquids than in gases.
 - Making sound waves require a vibrating source. Vibrating sources cause the air around them to vibrate. These vibrations are passed through the air to our ears where they cause the eardrum to vibrate and we hear sound.
 - Sonar is used to measure the depth of water.
 - A pulse of ultrasound is sent down from the boat and reflects from the seabed. Time taken for reflected pulse to reach boat is measured. This is used to calculate the speed of sound in water.
-

13. Light

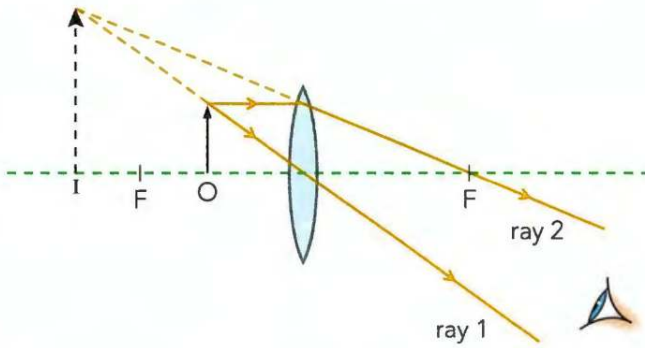
- **Normal** → the line drawn at right angles to a surface at the point where a ray hits the surface.
- **Angle of incidence** → the angle between the incident ray and the normal drawn at the point where the ray hits the surface.
- **Angle of reflection** → the angle between the reflected ray and the normal drawn at the point where the ray hits the surface.
- **Angle of refraction** → the angle between a refracted ray and the normal to the surface at the point where it passes from one medium to another.
- **Critical angle** → the minimum angle of incidence at which total internal reflection occurs.

- **Refractive index (n)** → is the ratio of the speeds of a wave in two different regions.
- Characteristics of image formed by a plane mirror → same size, same distance from the mirror, virtual.
- For reflection, angle of incidence = angle of reflection.
- $n = \sin(i)/\sin(r)$
- $n = 1/\sin(c)$
- In optic fibers, signals/ messages are encoded in pulses. Pulses travel down the optic fiber by total internal reflection. Pulses are received at the other end. They are decoded.

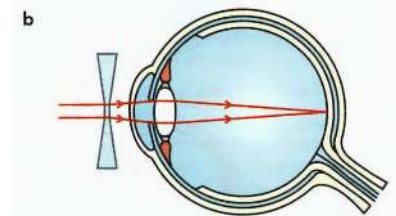
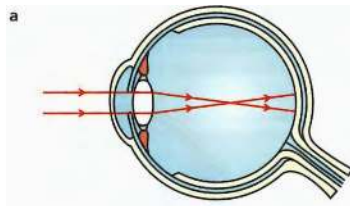
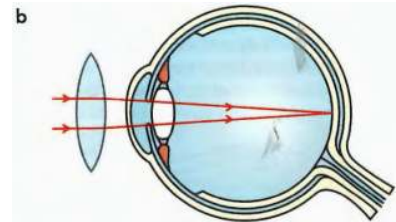
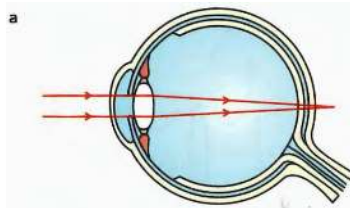
-
- Converging lens are fatter in the middle than at the edges.
 - Diverging lens are thinner in the middle than at the edges.



- **principal axis** → the line passing through the centre of a lens perpendicular to its surface
- **principal focus/focal point** → the point on the principal axis where rays of light parallel to the principal axis converge after passing through a converging lens
- **focal length** → the distance from the centre of the lens to its principal focus
- A virtual image is formed when diverging rays are extrapolated backwards and does not form a visible projection on a screen.
- Diagram of magnifying lens:



- Long sightedness is when near objects are blurry. Image is formed behind the retina. Converging lens are used.
- Short sightedness is when far objects are blurry. Image is formed in front of the retina. Diverging lens are used.

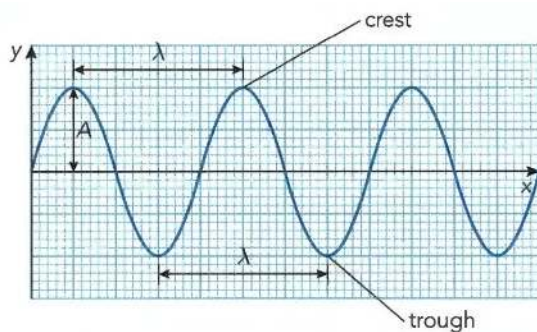


- Dispersion occurs because each colour is refracted by a different amount.
- Violet light slows down the most, and so it is refracted the most. Red light is least affected.
- Red travels fastest in a prism. Violet travel slowest in prism.
- In a prism, frequency remain constant, wavelength decrease, speed decrease, so dispersion occurs.
- Colours of rainbow in order from increasing frequency and decreasing wavelength
→ Red (Richard), Orange (Of), Yellow (York), Green (Gave), Blue (Battle), Indigo (In), Violet (Vain).

- Visible light of a single frequency is described as monochromatic.
-

14. Properties of Waves

- Waves transfer energy without transferring the matter itself.
- Wave motion illustrated by spring is longitudinal waves. Wave motion illustrated by rope is transverse wave. Water waves are also transverse waves.



- $v = f \lambda$
 - For a transverse wave, the direction of vibration is at right angles to the direction of propagation and the electromagnetic radiation, water waves and seismic S-waves (secondary) can be modelled as transverse.
 - For a longitudinal wave, the direction of vibration is parallel to the direction of propagation and the sound waves and seismic P-waves (primary) can be modelled as longitudinal
 - If the gap size = wavelength, then diffraction is maximum. If gap size increase and wavelength is constant then less diffraction and vice versa.
 - If the wavelength increase then there is more diffraction
-

15. The electromagnetic spectrum

- Roman(Radio) → men(Microwave) → invented(Infrared) → very(Visible) → unusual(Ultraviolet → x-ray(X-ray) → guns(Gamma rays). This is in order of

decreasing wavelength and increasing frequency.

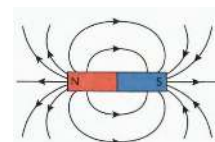
- Electromagnetic waves travel at the same speed in vacuum and air.
- Use of different regions of electromagnetic spectrum :
 - **radio waves:** radio and television transmissions, astronomy, radio frequency identification (RFID)
 - **microwaves:** satellite television, mobile phones (cell phones), microwave ovens
 - **infrared:** electric grills, short range communications such as remote controllers for televisions, intruder alarms, thermal imaging, optical fibres
 - **visible light:** vision, photography, illumination
 - **ultraviolet:** security marking, detecting fake bank notes, sterilising water
 - **X-rays:** medical scanning, security scanners
 - **gamma rays:** sterilising food and medical equipment, detection of cancer and its treatment
- Harmful effects on people of excessive exposure to electromagnetic radiation, including:
 - **microwaves:** internal heating of body cells
 - **infrared:** skin burns
 - **ultraviolet:** damage to surface cells and eyes, leading to skin cancer and eye conditions
 - **X-rays and gamma rays:** mutation or damage to cells in the body
- Communication with artificial satellites is mainly by microwaves:
 - some satellite phones use low orbit artificial satellites
 - some satellite phones and direct broadcast satellite television use geostationary satellites
- Mobile phones (cell phones) and wireless internet use microwaves because microwaves can penetrate some walls and only require a short aerial for transmission and reception

- Bluetooth uses radio waves because radio waves pass through walls but the signal is weakened on doing so.
- Optical fibres (visible light or infrared) are used for cable television and high-speed broadband because glass is transparent to visible light and some infrared; visible light and short wavelength infrared can carry high rates of data.
- Sound can be transmitted as a digital or analogue signal.
- Benefits of digital signaling:
 - increased rate of transmission of data
 - increased range due to accurate signal regeneration

Difference between Analog and Digital Signal	
Analog Signals	Digital Signals
Continuous signals	Discrete signals
Represented by sine waves	Represented by square waves
Human voice, natural sound, analog electronic devices are a few examples	Computers, optical drives, and other electronic devices
Continuous range of values	Discontinuous values
Records sound waves as they are	Converts into a binary waveform
Only used in analog devices	Suited for digital electronics like computers, mobiles and more

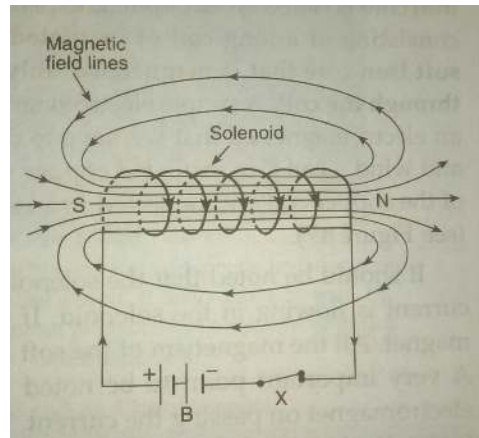
16. Magnetism

- **Induced magnetism** → The process where any unmagnetized object gains temporary magnetic capabilities when it is put inside the magnetic field of any magnet.



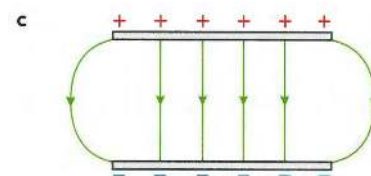
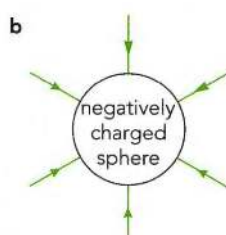
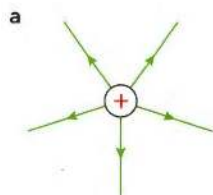
Type of magnetic material	Description	Examples	Uses
hard	retains magnetism well, but difficult to magnetise in the first place	hard steel	permanent magnets, compass needles, loudspeaker magnets
soft	easy to magnetise, but readily loses its magnetism	soft iron	cores for electromagnets (discussed later in the chapter), transformers and radio aerials

- Substances that are attracted by a magnet are called magnetic substances.
Example: Iron, cobalt, nickel, etc.
- Substances that are not attracted by a magnet are called non-magnetic materials.
Example: Aluminium, copper, wood, etc.
- Magnetic field is a region in which a magnetic pole experiences a force.
- The direction of a magnetic field at a point is the direction of the force on the N pole of a magnet at that point.
- **To plot magnetic field around bar magnet** → Place a magnet under a stiff sheet of plain paper or (preferably) clear plastic. Sprinkle filings over the paper or plastic. Tap the paper or plastic to allow the filings to move slightly so that they line up in the field. Draw the magnetic field lines. use a plotting compass to know the direction of the magnetic field. The compass points in the direction of the magnetic field.
- Magnetic forces are due to interactions between magnetic fields. When two bar magnets are placed close together their magnetic fields interact (affect each other) and produce a new pattern of magnetic lines of force. From these patterns, it is possible to say whether the magnets will attract or repel.
- The relative strength of a magnetic field is represented by the spacing of the magnetic field lines. The closer the field lines the stronger the magnetic field.
- To increase the strength of a electromagnet:
 - add more turns to the coil
 - increase the current
 - use a soft iron core



17. Static electricity

- Like charges repel and unlike charges attract.
- Charging of solids by friction involves only a transfer of negative charge (electrons). Positive charges (protons) do not move.
- In electrical conductors mobile electrons can move through the solids and carry the current. In electrical non conductors electrons can not move and cannot carry the current.
- Charge is measured in coulombs.
- Electric field is a region in which an electric charge experiences a force.
- The direction of an electric field at a point is the direction of the force on a positive charge at that point.



18. Electrical quantities

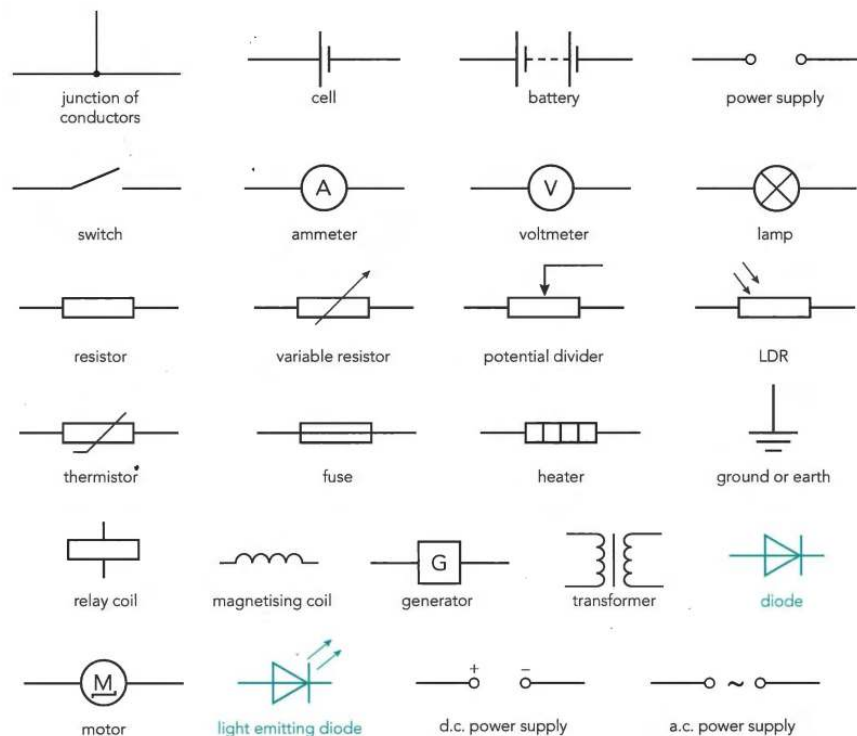
- Electric current is related to the flow of charge.
- Ammeters are connected in series.
- In metals electrons can move through the lattice and carry the electrical current.

Alternating Current	Direct Current
AC is easy to be transferred over longer distances – even between two cities – without much energy loss.	DC cannot be transferred over a very long distance. It loses electric power.
The rotating magnets cause the change in direction of electric flow.	The steady magnetism makes DC flow in a single direction.
The frequency of AC is dependent upon the country. But, generally, the frequency is 50 Hz or 60 Hz.	DC has no frequency or zero frequency.
In AC the flow of current changes its direction forward and backward periodically.	It flows in a single direction steadily.
Electrons in AC keep changing their directions – backward and forward.	Electrons only move in one direction – forward.

- **Definition** → Electric current is the charge passing a point per unit time; $I = Q/t$
 - Conventional current is from positive to negative and that the flow of free electrons is from negative to positive.
-
- **Definition** → electromotive force (e.m.f.) is the electrical work done by a source in moving a unit charge around a complete circuit.
 - e.m.f. is measured in volts (V).
 - **Definition** → potential difference (p.d.) is the work done by a unit charge passing through a component.
 - p.d. between two points is measured in volts (V).
 - Voltmeter is used in parallel with the component.
 - $E = W/Q$ e.m.f = work done/ charge
 - $V = W/Q$ p.d = work done/ charge
-

- $R = V/I$
- resistance is directly proportional to length
- resistance is inversely proportional to cross-sectional area
- In a filament lamp, at first, when the voltage and current are small, the lamp behaves like an ohmic resistor. However, as the voltage increases, the current causes the filament to get hot and glow brightly. At high temperatures, the filament has a higher resistance and so the current does not increase as rapidly as it would do if the filament had remained cool.
- electric circuits transfer energy from a source of electrical energy, such as an electrical cell or mains supply, to the circuit components and then into the surroundings.
- $P = IV$
- $E = IVt$

19. Electrical circuits



- Current at every point in a series circuit is the same.
- To calculate the combined emf of a source just add them up.
- To calculate the combined resistance of a circuit in series just add them up.
- For a parallel circuit, the current from the source is larger than the current in each branch.
- The combined resistance of two resistors in parallel is less than that of either resistor by itself.
- Advantage of connecting lamps in parallel:
 - if one lamp fail the other lamps will continue working.
 - every lamp can have its own switch.
- The sum of the currents entering a junction in a parallel circuit is equal to the sum of the currents that leave the junction because current is flow of electrons and electrons can not just disappear in air.
- The total p.d. across the components in a series circuit is equal to the sum of the individual p.d's across each component.
- The p.d. across an arrangement of parallel resistances is the same as the p.d. across one branch in the arrangement of the parallel resistances.
- Resistance in parallel = $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$
- The p.d. across an electrical conductor increases as its resistance increases for a constant current.
- The equation for two resistors used as a potential divider

$$\frac{R_1}{R_2} = \frac{V_1}{V_2}$$

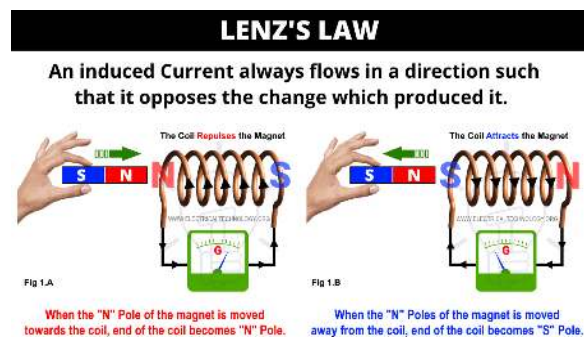
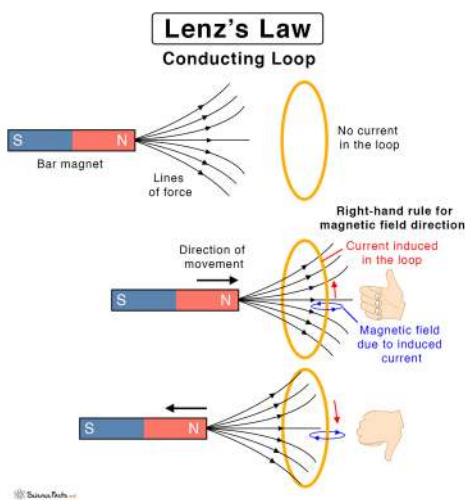
-
- When insulation is damaged, exposed metal parts may become energized if live wires contact one another.

- If an excessive current flows in a wires. It will overheat and the insulation may melt, causing it to emit poisonous fumes or even catch fire.
 - In damp conditions the water can conduct electricity. If your are in contact you may provide a conductive path for current to flow from a live wire through you to earth. This could prove fatal.
 - Block adapters do not usually contain fuse so there is an increased danger that a current will exceed the rating of the wall socket. Overloading the socket in this way increases the chance that it will draw excess current, and the plug, will heat up and catch fire.
 - A mains circuit consists of a live wire (line wire), a neutral wire and an earth wire.
 - A switch on the electrical appliance must be connected to the live wire. If it was connected to either the neutral wire or the earth wire, a current could still pass into the appliance even if the switch was open (off) and this could lead to a fire or to somebody being electrocuted if they touched a faulty appliance.
 - Fuse stop excess current flowing through a circuit. A fuse contains a thin section of wire, designed to melt and break if the current gets above a certain value. The thicker the wire used for the fuse, the higher the current that is needed to make it melt (or blow).
 - The current rating of the fuse should be just above the value of the current that flows when the appliance is operating normally.
 - A fuse without an earth wire protects the circuit and the cabling for a double-insulated appliance.
 - When the electric circuit for an electrical appliance is placed inside a case made from an electrical insulator so that it is impossible for a live wire to touch the outer casing so there is no need for earthing in this case.
-

20. Electromagnetic forces.

- A conductor moving across a magnetic field or a changing magnetic field linking with a conductor can induce an e.m.f. in the conductor.
- Factors that affect the magnitude of induced emf:

1. Magnetic strength of the core in the coil of wire. (stronger --> bigger emf)
 2. Number of turns of wire in the coil (more turns --> bigger emf)
 3. The cross-sectional area of the coil (bigger area --> bigger emf). (This could also be seen as the angle you point the magnet at the coil, because flux linkage is basically the same as flux cutting.)
 4. How fast you move the magnet into/out of the coil. (Faster --> bigger emf)
- The direction of an induced e.m.f. opposes the change causing it.



- To find the direction of magnetic fields in a straight current carrying conductor use the right hand grip rule.
- Inside the coil the field lines run parallel to each other showing that the field is uniform (its strength is constant)
- When field lines are close together the strength is stronger and vice versa.
- If current is increased field lines come closer, strength increase. If current is reversed the direction of magnetic field decrease.

Electromagnetism in action – the relay

A relay is a switch operated by an electromagnet. In Chapter 19, we saw how a relay can be used in an electric circuit. One type is shown in Figure 20.8, together with the circuit symbol.

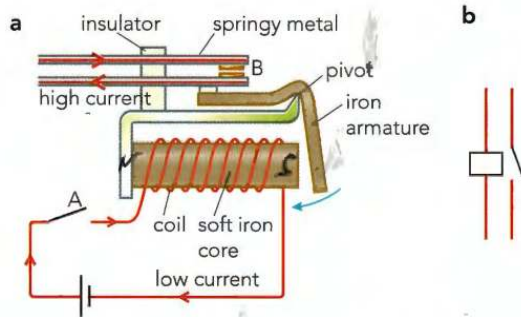
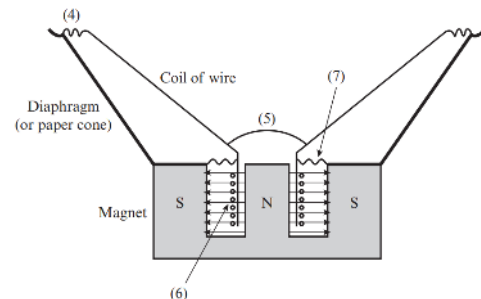


Figure 20.8a: A relay, capable of switching a circuit carrying hundreds of amps. **b:** The circuit symbol for a relay. The rectangle represents the electromagnet coil.

- When switch A is closed, a small current flows around the circuit through the coil of the electromagnet.
- The electromagnet attracts the iron **armature**. As the armature tips, it pushes the two contacts at B together, completing the second circuit.
- Notice that there is no electrical connection between the two circuits.

- A relay is used to make a small current switch a larger current on and off

- A current flows through the coil, which produces a magnetic field around the coil. This magnetic field interacts with the field between the poles of the (permanent) magnet. This produces a force, the direction of which is given by Fleming's left-hand rule. On the left, the force will be down the page. On the right, it will also be down the page (because, although the direction of the current flow is reversed, so is the direction of the magnetic field between the poles of the magnet).



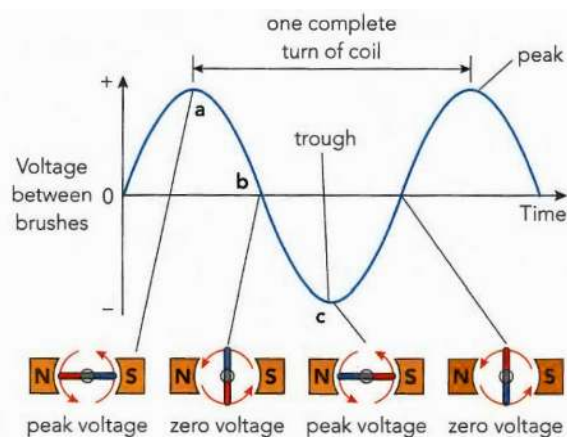
Therefore the coil moves down. The current through the coil is alternating. As the current changes direction, so does the force on each side of the coil. This makes the coil vibrate. This makes the diaphragm (attached to the coil) vibrate. This makes a series of compressions and

rarefactions in the air. Making a sound wave.

- Fleming's left hand rule can be used with motors.

21. Electromagnetic Induction

- As the coil rotates, each side of the coil passes first the magnetic north pole and then the south pole. This cuts the field lines. When the coil is horizontal, it cuts through the field lines inducing a voltage. As it turns to vertical, it cuts fewer field lines so the voltage decreases to zero. As it continues back to horizontal it cuts through the field lines in the opposite direction, giving a peak voltage in the opposite direction.
- Slip rings: a device used to allow current to flow to and from the coil of an a.c. generator



- Alpha particle deflects towards the negatively charged plate. Beta particle deflects towards the positively charged plate. Gamma rays do not deflect at all.
- Transformer number of turn equation

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

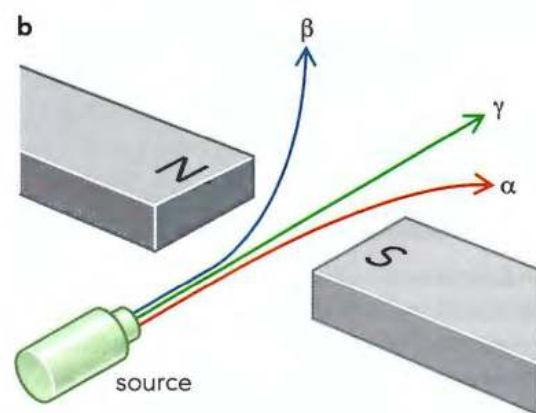
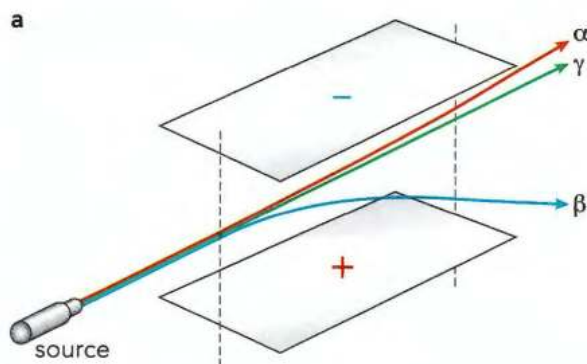
- Power loss equation $\rightarrow P = I^2 R$
 - $I_p V_p = I_s V_s$
 - As the voltage increase the current decrease. Using the power loss equation as the current decrease the power loss decrease.
-

22. The nuclear atom

- An atom in consist of a positively charged nucleus and negatively charged electrons in orbit around the nucleus.
 - Atoms may form positive ions by losing electrons or form negative ions by gaining electrons.
 - Evidence provided by scattering alpha particle gold foil experiment:
 - a very small nucleus surrounded by mostly empty space
 - a nucleus containing most of the mass of the atom
 - a nucleus that is positively charged
 - Nucleus of an atom contains protons and neutrons.
 - Relative charges of protons, neutrons and electrons are +1, 0 and -1 respectively.
 - Proton number $\rightarrow$ number of protons in the nucleus of an atom.
 - Nucleon number $\rightarrow$ number of proton + neutron in the nucleus of an atom.
 - Isotopes are the atoms of the same element, have the same proton number but different neutron/ nucleon number.
 - An element may have more than one isotope.
 - Nuclear fission is splitting of nuclei to release energy. Occur in nuclear reactor at earth.
 - Nuclear fusion is the joining of hydrogen nuclei to form helium nuclei and release energy. Occurs in the sun.
-

23. Radioactivity

- Background radiation → the radiation from the environment to which we are exposed all the time/ it is present all the time.
 - Sources of radiation:
 - a. radon gas (in the air)
 - b. rocks and buildings
 - c. food and drink
 - d. cosmic rays
 - Ionising nuclear radiation can be measured using a detector connected to a counter.
 - Count rate is measured in count / second or count / minutes.
 - Correct count rate due to a source = count rate - background radiation.
-
- Emission of radiation from a nucleus is spontaneous and random in direction.
 - Alpha are highly ionising. Beta are moderately ionising. Gamma are least ionising.
 - Alpha → $2+$; Beta → -1 ; Gamma → 0
 - Alpha is least penetrating. Beta is moderately penetrating. Gamma is highly penetrating.



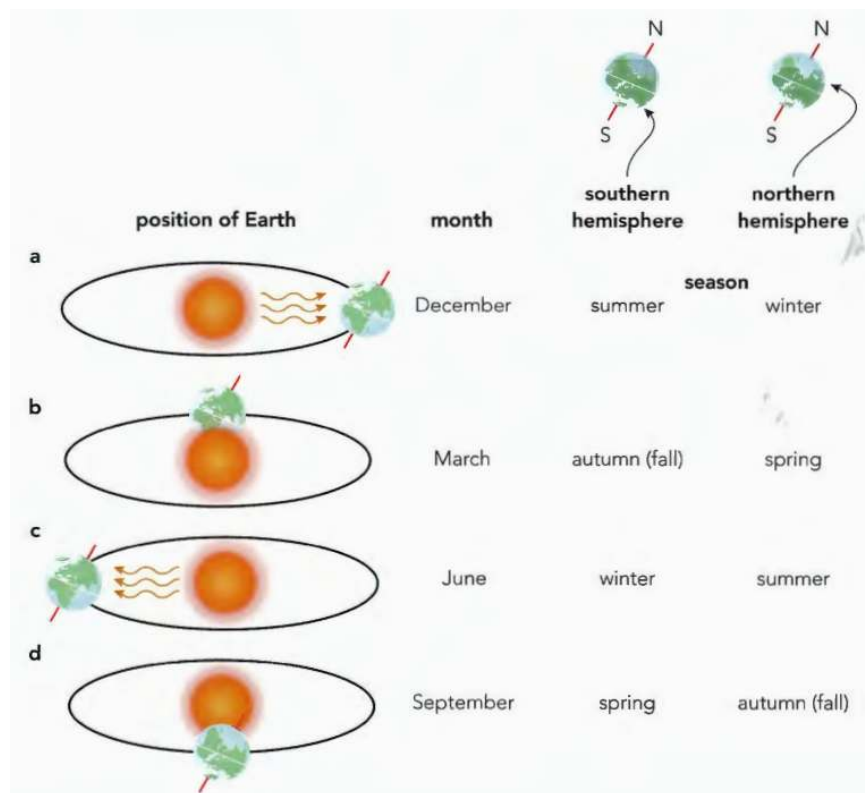
- Consider an alpha-particle passing through the air. An alpha-particle is the slowest moving of all the three radiations and has the largest charge. As the alpha-particle collides with an air molecule, it may knock an electron from the air molecule, so that it becomes charged. The alpha-particle loses a little of its energy. It must ionise thousands of molecules before it loses all of its energy and comes to a halt, alpha-radiation is the most strongly ionising radiation.
 - Radioactive decay is a change in an unstable nucleus that can result in the emission of α -particles or β -particles and/or γ -radiation and these changes are spontaneous and random.
 - During α -decay or β -decay, the nucleus changes to that of a different element
 - Isotopes of an element may be radioactive due to an excess of neutrons in the nucleus and/or the nucleus being too heavy.
 - After emission, the nucleus of the radioactive isotope becomes more stable and reduce in the number of excess electrons.
 - During β emission, neutron $\Rightarrow$ proton + electron.
 - Half life $\rightarrow$ the time taken for half the nuclei of that isotope in any sample to decay.
-
- Working of smoke alarm:
 - Radiation from the source falls on a detector. Since alpha-radiation is charged, a small current flows in the detector. The output from the processing circuit is off, so the alarm is silent.
 - When smoke enters the gap between the source and the detector, it absorbs the alpha radiation. Now no current flows in the detector, and the processing circuit switches on, sounding the alarm.
 - Alpha is chosen because it is easily absorbed by the smoke.
 - Thickness measurement:
 - Beta particles are used. Beta particles are directed through the material. The radiation is detected by the detector on the other end.
 - If the material is too thick, the radiation level will be low and an automatic control system adjusts the thickness.

- Beta-radiation is used in this application because alpha-radiation would be absorbed entirely by the paper, plastic or aluminium.
 - Gamma rays are used in fault detection in sewage pipes.
 - Gamma rays are used in cancer treatment. A source of γ -rays (or X-rays) is directed at the tumour that needs to be destroyed. The source moves around the patient, always aiming at the tumour. In this way, other tissues receive only a small dose of radiation.
 - Gamma rays are used in food irradiation. Intense gamma rays can kill single celled bacteria or organism in the food.
 - In sterilisation of medical equipment gamma rays are used. Equipment is packed in plastic bags and then exposed to gamma rays. gamma rays are used because it can pass through thick equipment making sure all parts are sterilised.
 - Gamma rays are used in diagnosis of some diseases too. Radioactive chemicals are taken in that release gamma rays. It is detected by detector. Higher count rate where there is a blockage of blood. Gamma rays are used because its source have a relatively short half life of about 6 hours, but this is long enough for the investigation.
-
- Ionising radiation can cause, death of cells, mutation and cancer.
 - Radioactive sources are kept in lead lined wooden boxes. lead absorbs all the radiation
 - Radiation suits are worn in radioactive areas.
 - Radiographers operate equipment from a separate room, to reduce exposure time.
 - Tweezers/ tongs are used to handle radioactive sources to increase distance between source and living tissue.
-

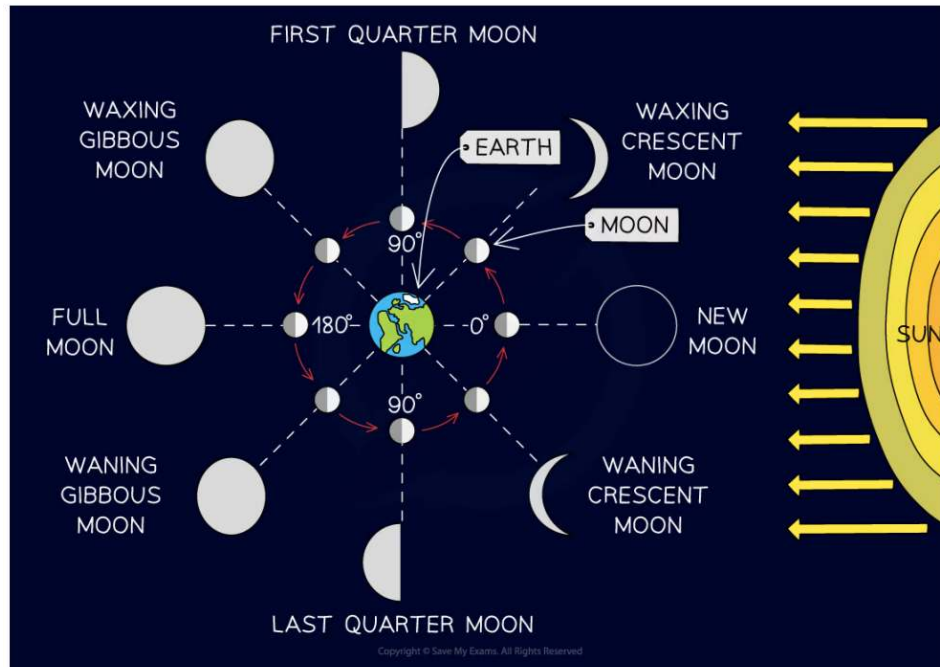
24. Earth and The Solar System

- Earth is a planet that rotates on its axis, which is tilted, once in approximately 24 hours.

- The side of the Earth facing the Sun experiences daylight whilst the other side is in darkness. As the Earth turns, Sun appears directly overhead at midday. As the Earth continues to turn, the spot moves out of the direct sunlight until, at sunset, the Sun appears to slip below the western horizon.
- Earth orbits the Sun once in approximately 365 days.
- Countries at the Equator do not experience seasons because the Sun's rays always hit them at the same angle.



- It takes approximately one month for the Moon to orbit the Earth.
- The Moon revolves around its own axis in a month so always has the same side facing the Earth at all times. We never see the hemisphere that is always facing away from Earth, although astronauts have orbited the Moon and satellite have photographed it.
- The Moon shines with reflected light from the Sun, it does not produce its own light.



- In the New Moon phase:
 - The Moon is between the Earth and the Sun. Therefore, the sunlight is only on the opposite face of the Moon to the Earth. This means the Moon is unlit as seen from Earth, so it is not visible.
- At the Full Moon phase:
 - The Earth is between the Moon and the Sun. The side of the Moon that is facing the Earth is completely lit by the sunlight. This means the Moon is fully lit as seen from Earth.
- In between, a crescent can be seen where the Moon is partially illuminated from sunlight.
- Average orbital speed = $(2\pi \times \text{avg. radius}) / \text{Time}$
- $v = 2\pi r/T$
- Solar System contains:
 - one star, the Sun
 - eight planets, Mercury (My), Venus (Very), Earth(Elegant), Mars (Mother), Jupiter (Just), Saturn (Served), Uranus (Us), Neptune (Noodles).

- minor dwarf planets, like Pluto and Eris.
 - asteroids in asteroids belts between Mars and Jupiter.
 - moons, that orbit the planets.
 - comets, which are giant snowballs, orbit the Sun in very irregular orbits. When they are furthest from the Sun, they are frozen balls of gas, rock and dust. As they get near the Sun they heat up, ice melts and leave a trail of dust and gas behind them. They have highly elliptical orbits.
- Minor planets and comets have elliptical orbits, and the Sun is not at the centre of the elliptical orbit, except when the orbit is approximately circular.
 - The four planets nearest to the Sun are rocky and small and the four planets furthest from the Sun are gaseous and large.
 - The Solar System began as a nebula, which is a huge swirling ball of dust and gas. Most of this gas was hydrogen, but there were also other elements formed by fusion in other stars, which had exploded at the end of their life cycle, sending their contents out into the clouds of interstellar gas. As gravity pulled this mass together, the centre formed a star. The planets formed from the materials of the nebula which were not pulled into the Sun. The spinning motion of the dust and gas formed a flat, spinning ring disc known as an accretion disc. Gravity pulled dust and gas together so they joined to make rocks which then join to make larger rocks. The process of the dust and gas being pulled together by gravity is called accretion and it led to the formation of the inner, rocky planets. The intense heat forced some of the lighter materials further away and these formed the outer planets-the gas giants.
 - Orbital distance is the distance from from the sun to the planet
 - Orbital duration is the time taken for a planet to complete one revolution around the sun relative to earth. Earths orbital duration is 1 year, Jupiter's orbital duration is 12 years.
 - Strength of the gravitational field at the surface of a planet depends on the mass of the planet.
 - Strength of the gravitational field around a planet decreases as the distance from the planet increases.

- The Sun contains most of the mass of the Solar System and this explains why the planets orbit the Sun.
 - The force that keeps an object in orbit around the Sun is the gravitational attraction of the Sun.
 - The strength of the Sun's gravitational field decreases and the orbital speeds of the planets decrease as the distance from the Sun increases.
 - An object in an elliptical orbit travels faster when closer to the Sun because when an orbiting object is at its closest to the Sun, it has its maximum kinetic energy and minimum gravitational potential energy. When it is furthest from sun it has its maximum gravitational potential energy and minimum kinetic energy.
 - Speed of light is $3.0 \times 10^8 m/s$.
-

25. Stars and The Universe

- Sun is a star of medium size, consisting mostly of hydrogen and helium, and it radiates most of its energy in the infrared, visible and ultraviolet regions of the electromagnetic spectrum.
 - Stars are powered by nuclear reactions that release energy and in stable stars the nuclear reactions involve the fusion of hydrogen into helium.
-
- Galaxies are each made up of many billions of stars.
 - The Sun is a star in the galaxy known as the Milky Way.
 - Other stars that make up the Milky Way are much further away from the Earth than the Sun is from the Earth.
 - Astronomical distances can be measured in light-years, where one light-year is the distance travelled in (the vacuum of) space by light in one year.
 - one light-year is equal to $9.5 \times 10^{15}m$.
 - **Life cycle of a Star :**
 - a star is formed from interstellar clouds of gas and dust that contain hydrogen.

- a protostar is an interstellar cloud collapsing and increasing in temperature as a result of its internal gravitational attraction.
 - a protostar becomes a stable star when the inward force of gravitational attraction is balanced by an outward force due to the high temperature in the centre of the star.
 - all stars eventually run out of hydrogen as fuel for the nuclear reaction.
 - most stars (mass ≤ 8 solar masses) expand to form red giants and more massive stars (mass > 8 solar masses) expand to form red super-giants when most of the hydrogen in the centre of the star has been converted to helium.
 - a red giant from a less massive star forms a planetary nebula with a white dwarf star at its centre. then this white dwarf forms a black dwarf.
 - a red supergiant explodes as a supernova, forming a nebula containing hydrogen and new heavier elements, leaving behind a neutron star or a black hole at its centre.
 - the nebula from a supernova may form new stars with orbiting planet.
-
- Milky Way is one of many billions of galaxies making up the Universe and the diameter of the Milky Way is approximately 100000 light-years.
 - Redshift is an increase in the observed wavelength of electromagnetic radiation emitted from receding stars and galaxies.
 - The light emitted from distant galaxies appears redshifted in comparison with light emitted on the Earth.
 - Redshift in the light from distant galaxies is evidence that the Universe is expanding and supports the Big Bang Theory.
 - Microwave radiation of a specific frequency is observed at all points in space around us and is known as cosmic microwave background radiation (CMBR).
 - The CMBR was produced shortly after the Universe was formed and this radiation has been expanded into the microwave region of the electromagnetic spectrum as the Universe expanded.
 - The speed v at which a galaxy is moving away from the Earth can be found from the change in wavelength of the galaxy's starlight due to redshift.

- The distance of a far galaxy d can be determined using the brightness of a supernova in that galaxy.
- **Hubble constant** H_0 is the ratio of the speed at which the galaxy is moving away from the Earth to its distance from the Earth.
- $H_0 = v/d$
- The current estimate for H_0 is 2.2×10^{-18} per second.
- The equation: $d/v = 1/H_0$, represents an estimate for the age of the Universe and that this is evidence for the idea that all the matter in the Universe was present at a single point.

- When current pass through galvanometer it shows deflection in the direction of convectional current.
- Circuit breaker, earth wire and fuse are all designed to improve the safe working of a mains electrical supply.
- Decreasing the surface area of an object in liquid doesn't increase or decrease the pressure of liquid on that object.
- For TIR to take place light must travel from denser medium to rarer medium.

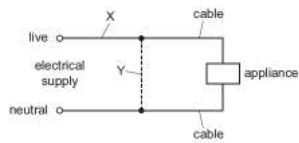
20 In a cold country, a bicycle has been left outside all night. The cyclist finds the plastic hand grips feel less cold to the touch than the steel handlebars.

Which row correctly describes the temperature and the property of the two materials?

	the temperature of the two materials	the property of the two materials
A	the temperature of the steel is much lower than that of the plastic	the plastic is a better thermal conductor than the steel
B	the temperature of the steel is much lower than that of the plastic	the steel is a better thermal conductor than the plastic
C	the steel and the plastic are both at the same temperature	the plastic is a better thermal conductor than the steel
D	the steel and the plastic are both at the same temperature	the steel is a better thermal conductor than the plastic

- Both fuse and circuit breakers are connected in series.

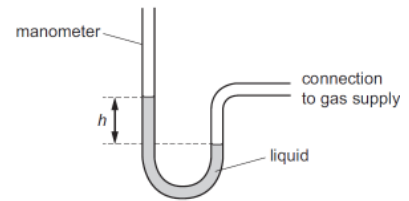
- 34 Either a fuse or a circuit-breaker can be used to protect electrical cables from large currents that could cause overheating.



When a fuse is used, where should it be connected, and when a circuit-breaker is used, where should it be connected?

	position of fuse	position of circuit-breaker
A	X	X
B	X	Y
C	Y	X
D	Y	Y

- 12 A manometer is used to measure the pressure of a gas supply.

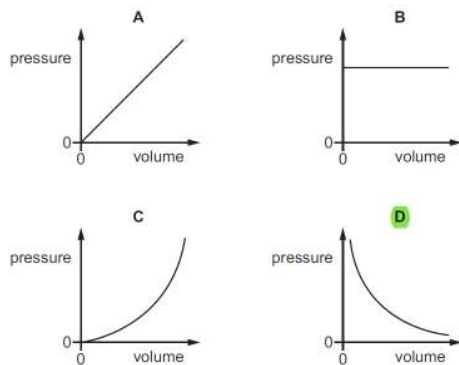


Which change gives a greater value of height h ?

- A** using a less dense liquid
- B** using a more dense liquid
- C** using a narrower tube
- D** using a wider tube

- 15 The gas in a sealed container is compressed at constant temperature.

Which graph shows how the pressure of the gas changes with its volume?



- 17 The diagram shows a liquid-in-glass thermometer.



Which change increases the sensitivity of the thermometer?

- A** a narrower capillary tube
- B** a wider capillary tube
- C** thicker glass around the bulb
- D** thinner glass around the bulb

- 19 On a cold day, a shiny metal rod feels colder to the touch than a black plastic rod.

Which statement explains this observation?

- A** The metal rod is a better absorber of infra-red radiation than the plastic rod.
- B** The metal rod is a better thermal conductor than the plastic rod.
- C** The metal rod is a worse absorber of infra-red radiation than the plastic rod.
- D** The metal rod is a worse thermal conductor than the plastic rod.

- Chemical energy is stored in batteries which is then converted to electrical energy.
- Electrical shock is the greatest hazard from uninsulated wires.
- Fuse and switches are both connected to the live wire.
- Speed of wave does not change during diffraction.